Lecture 31.07: K-Means Clustering — Scikit-Learn Implementation











📌 Topic Overview

Practical implementation of K-Means clustering using scikit-learn. Covers synthetic cluster generation with make_blobs, KMeans instantiation, fitting, extracting cluster_centers_, and visualizing colored cluster partitions.

📋 Key Learning Objectives

- Generating synthetic 2D isotropic Gaussian blobs with make_blobs.
- Importing and configuring KMeans from sklearn.cluster.
- Hyperparameters: n_clusters=4, init='k-means++', max_iter=300, random_state=42.
- Executing fit_predict(X) to obtain cluster labels.
- Extracting and plotting cluster_centers_ with Seaborn/Matplotlib.

🔍 In-Depth Conceptual Breakdown

- **Data Generation**: `make_blobs(n_samples=500, centers=4, n_features=2, cluster_std=1.0, random_state=42)` use karke 4 distinct clusters banaye gaye.
- **KMeans Model**: `kmeans = KMeans(n_clusters=4, init='k-means++', max_iter=300, random_state=42)` instantiate kiya gaya.
- **Fitting & Prediction**: `labels = kmeans.fit_predict(X)` execute karke har sample ko `0, 1, 2, 3` cluster ID assign hui.
- **Centroids Extraction**: `centroids = kmeans.cluster_centers_` 4 coordinates return karta hai jo har cluster ka mathematical center of mass hain.

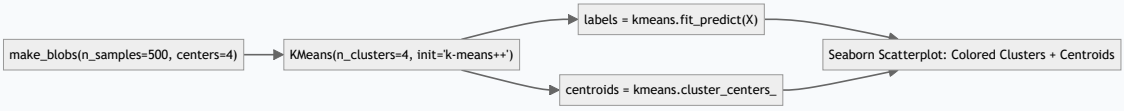
- **Visualization**: Seaborn `scatterplot` se points ko unke cluster label ke hisab se color kiya gaya aur centroids ko red markers se plot kiya gaya.
- **Code Lecture**: Is lecture mein ma'am ne notebook run karke sikhaya hai, isliye iska complete code lecture_31_07.ipynb mein available hai.

📋 KMeans Key Hyperparameters in Scikit-Learn

Hyperparameter	Default Value	Recommended Setting	Description
`n_clusters`	`8`	Domain-specific / Elbow K	The number of clusters and centroids to form
`init`	`'k-means++'`	`'k-means++'`	Centroid initialization strategy to avoid traps
`n_init`	`'auto'` (10)	`10` or `'auto'`	Number of times K-Means will run with different seeds
`max_iter`	`300`	`300`	Maximum iterations for a single run
`tol`	`1e-4`	`1e-4`	Relative tolerance with regards to Frobenius norm to declare convergence

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📊 Scikit-Learn K-Means Execution Flow



🔢 Mathematical Formulation

Cluster assignment mapping for sample i:

$$y_i = \arg \min_{k \in \{1,\dots,K\}} \|\mathbf{x}_i - \mathbf{C}_k\|^2$$

💻 Implementation

The standalone verified code is provided in the synchronized Jupyter Notebook.